

S GOKUL KRISHNA

Platform Engineer | Backend Developer (Python / FastAPI / PostgreSQL)

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SUMMARY

Platform Engineer and Backend Developer with 2 years of experience building and operating production backend systems, databases, and infrastructure across on-premise and cloud environments. Hands-on expertise in Python (**FastAPI**), **PostgreSQL**, REST API design, **Docker**, **CI/CD**, and **Linux** systems administration, spanning multi-VM production deployments, observability, and enterprise (**SAP**) integrations. Track record of resolving cross-stack production incidents — database performance, load-balancer routing, service-worker caching, container networking — for a multi-site industrial client running across two data centers.

EXPERIENCE

Platform Engineer & Backend Developer — Lumbini Elite Solutions, Bangalore Sep 2024 – Present

- Built and operate backend services and **REST APIs** in Python **FastAPI** for **Petronet CDP**, a production application deployed across a multi-VM architecture spanning Mumbai and Chennai data centers, serving concurrent enterprise users.
- Diagnosed and resolved a production concurrency ceiling (100 concurrent users), scaling the system to reliably support 250+ concurrent users by re-tuning **PostgreSQL** connection pooling (max_connections, shared_buffers, work_mem), redesigning per-environment app connection pools, and staggering **APScheduler cron** jobs across nodes.
- Debugged a geo-fencing/location-access-control failure spanning frontend (PWA service-worker cache bypass) and infrastructure (**OpenStack Octavia** load balancer not forwarding client IPs), restoring accurate station-matching logic in production.
- Designed relational **PostgreSQL** schemas, wrote stored procedures/functions, and managed schema evolution via **Alembic** migrations, including a multi-table migration spanning safety checklists, work permits, logbooks, and HR/leave modules.
- Deployed a full observability stack (**Prometheus**, **Grafana**, **Loki**, **Alertmanager**, Blackbox/Node/Postgres/**Nginx** exporters) across 10+ VMs with custom app metrics and **AWS SES** alerting.
- Designed and built an automated cross-data-center backup architecture replicating production data from Mumbai to a Chennai **NFS** share over **SSH** (incremental **rsync**), scheduled via **cron** with weekly **logrotate** (4-rotation retention), providing disaster-recovery coverage for the multi-VM **Petronet CDP** deployment.
- Implemented **CI/CD** pipelines with **GitHub Actions** using self-hosted runners, driving tag-triggered production deployments across frontend and backend services with automated health checks and database migrations.
- Automated 250+ GB of data migration using **Python Selenium** pipelines, cutting manual data-entry effort by over 90%.
- Delivered enterprise automation workflows integrating with **SAP** systems for material creation, and provided technical advisory input on an **SAP NetWeaver-to-BTP** migration strategy that eliminated the two largest projected cost drivers.
- Configured **Nginx** reverse proxies (SSL/TLS termination, access control) and deployed **Apache Superset** BI, embedding dashboards into **React** apps; provisioned and hardened **Hyper-V Linux** infrastructure with self-healing network automation.
- Set up and deployed infrastructure for an Energy Management System (EMS) application using **Docker/Docker Compose**, with **Redis** for caching and **RabbitMQ** for task/message queuing.
- Leveraged agentic AI coding tools (**Claude Code**, **Google Antigravity**) and **VS Code** with structured prompt engineering to accelerate feature development, debugging, and documentation across the **FastAPI** backend and infrastructure codebase.
- Provisioned and configured **Linux** servers from scratch (OS install, user/**SSH** hardening, firewall, package/dependency setup, **Nginx**, **Docker**) for new environments, bringing each node to production-ready state before app deployment.

PROJECTS

Instacart BI Agent v2 — Conversational Analytics (Python, FastAPI-style guardrails, DuckDB, Groq/LLaMA 3.3, Streamlit, Plotly) | Live Demo

- Built a production-grade natural-language-to-SQL BI agent over a 33M+ row Instacart dataset, combining an LLM SQL-generation layer with a statistical guardrail layer (bias-correcting HAVING filters, correct chart-type enforcement, SQL validation) and an auto-repair loop that retries and fixes failed SQL automatically.
- Used **DuckDB** for in-memory columnar/vectorized query execution across orders, order_products, products, aisles, and departments tables, with a unified view combining prior and train order data; visualized results with **Plotly** and served the chat interface via **Streamlit**.

Chiller Fault Detection & Diagnosis (Python, 1D CNN) | github.com/gokulkrishna15/chiller_fdd

- Built a 1D **CNN** model and an **ANN + PSO** optimization model for industrial fault detection and diagnosis in chiller systems.

Additional projects: Real Estate Platform (**MERN** stack, full-stack auth & search — live demo) · Bird Species Classifier (**CNN**, 95% accuracy, deployed via **Streamlit**) · Fish Farming Decision-Support Tool (Python)

TECHNICAL SKILLS

- **Languages & Backend:** Python, C, SQL, FastAPI, REST API Design, Backend Architecture
- **Database:** PostgreSQL, Database Design, Stored Procedures & Functions, pgAdmin, Alembic Migrations, Connection-Pool & Query Optimization
- **DevOps & Infrastructure:** Docker, Docker Compose, CI/CD (GitHub Actions, self-hosted runners), Nginx, SSL/TLS, Linux (Ubuntu, CentOS), Hyper-V, OpenStack, Server Setup & Provisioning (bare OS to production, from scratch)
- **Messaging & Caching:** Redis, RabbitMQ
- **Observability:** Prometheus, Grafana, Loki, Alertmanager, Blackbox/Node/Postgres Exporters
- **Data & Automation:** Python Selenium, ETL Pipelines, SAP Automation, Enterprise Workflow Automation
- **Data Science & ML:** NumPy, Pandas, Matplotlib, CNN, ANN, PSO
- **BI & Visualization:** Apache Superset, Dashboard Development, DuckDB, Plotly, Streamlit Model Deployment
- **AI & Agentic Coding Tools:** Claude (Claude Code), Google Antigravity, VS Code (AI-assisted development), Prompt Engineering

EDUCATION

- PES University, Bangalore — B.Tech in Computer Science, CGPA: 7.33 (2020 – 2024)

CERTIFICATIONS

- Cloud Computing 101 — AWS Training ([Credential](#))
- **Linux** Kernel Development (LFD103) — **Linux** Foundation ([Credential](#))